

A BEAUTIFUL SOLUTION TO THE BASEL PROBLEM

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PREMABLE

The Basel Problem has long been hailed as a monumental link between analysis and geometry. The formula

$$(1) \quad \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots = \frac{\pi^2}{6}$$

is striking in that it relates the constant π , something which arises from the geometry of a circle, to the infinite sum $\sum_{n=1}^{\infty} \frac{1}{n^2}$, which seemingly has nothing to do with plane geometry. One can visualise this sum by considering the graph of $f(x) = \frac{1}{x^2}$ over the positive real line. Equation 1 tells us that if we consider all the points corresponding to natural inputs into f , the sum of the y -coordinates of those points in the output space somehow approaches $\frac{\pi^2}{6}$!

This seems like an impossible result to prove, at least simply. However, the tools developed in Fourier analysis render the proof into all but a wonderful interplay between ideas.

The proof I display in this paper has been inspired by my solution to Exercise 6, Chapter 2 of [SS03]. An explicit formula for the values of the Riemann zeta function at positive even integers is discussed in Exercise 19, Chapter 5 of this reference.

1. AN INTERESTING FUNCTION

Define a function f on the circle by $f(\theta) = |\theta|$, over the interval $[-\pi, \pi]$.

The Fourier coefficients are given by the integral

$$(2) \quad \hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta) e^{-in\theta} d\theta = \frac{1}{2\pi} \int_{-\pi}^{\pi} |\theta| e^{-in\theta} d\theta.$$

Now, for $n = 0$

$$\hat{f}(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} |\theta| d\theta = \frac{1}{\pi} \int_0^{\pi} \theta d\theta = \frac{\pi}{2}$$

Since f is an even function, the function's Fourier series will be a cosine series, so we may neglect the imaginary term of $e^{-in\theta}$ arising from Euler's Formula.

Equation (2) then simplifies to yield

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$$\hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} |\theta| e^{-in\theta} d\theta = \frac{1}{\pi} \int_{-\pi}^{\pi} \theta \cos n\theta d\theta.$$

In the case where $n \neq 0$, we may carry out integration by parts on our expression above:

$$\begin{aligned} \hat{f}(n) &= \frac{1}{\pi} \left(\left[\frac{1}{n} \sin n\theta \right]_0^{\pi} - \frac{1}{n} \int_0^{\pi} \sin n\theta d\theta \right) \\ &= -\frac{1}{\pi n} \int_0^{\pi} \sin n\theta d\theta \\ &= \frac{1}{\pi n^2} ((-1)^n - 1). \end{aligned}$$

We may therefore conclude that

$$(3) \quad \hat{f}(n) = \begin{cases} \frac{\pi}{2}, & n = 0 \\ \frac{1}{\pi n^2} ((-1)^n - 1), & n \neq 0. \end{cases}$$

2. MAIN ARGUMENT

We can now use our calculations in 1; namely Equation 3 to proceed with the main proof.

The Weierstrass M-Test [Abb15] shows that the series $\sum_{n=-\infty}^{\infty} |\hat{f}(n)|$ is absolutely (and in fact, uniformly) convergent, so by Chapter 2, Corollary 2.3 of [SS03], we have

$$(4) \quad f(\theta) = \frac{\pi}{2} + 2 \sum_{n=1}^{\infty} \frac{-1 + (-1)^n}{\pi n^2} \cos n\theta.$$

Setting $\theta = 0$ gives

$$(5) \quad 2 \sum_{n=1}^{\infty} \frac{-1 + (-1)^n}{\pi n^2} \cos n\theta = -\frac{\pi}{2}.$$

Note that

$$-1 + (-1)^n = \begin{cases} -2, & n \text{ odd} \\ 0, & n \text{ even.} \end{cases}$$

Using this result in Equation 5,

$$(6) \quad \sum_{n \text{ odd} \geq 1} \frac{1}{n^2} = \frac{\pi^2}{8}.$$

The equation above seems to indicate that we are stuck. After all, we have only calculated the sum for *odd* positive integers. But here's a beautiful trick:

$$(7) \quad \sum_{n=1}^{\infty} \frac{1}{n^2} = \sum_{n \text{ odd} \geq 1} \frac{1}{n^2} + \sum_{n \text{ even} \geq 1} \frac{1}{n^2}.$$

Since positive even numbers are of the form $2k$ for some $k \in \mathbb{Z}^+$, we may write the second sum as

$$\sum_{n \text{ even} \geq 1} \frac{1}{n^2} = \sum_{k=1}^{\infty} \frac{1}{(2k)^2} = \frac{1}{4} \sum_{k=1}^{\infty} \frac{1}{k^2}.$$

Using this result in Equation 7, along with Equation (6) yields

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{8} + \frac{1}{4} \sum_{k=1}^{\infty} \frac{1}{k^2}.$$

In particular, rearranging gives

$$\frac{3}{4} \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{8},$$

so that

$$(8) \quad \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6},$$

which proves the Basel Problem!

REFERENCES

- [Abb15] Stephen Abbott. *Understanding Analysis*. 2nd ed. Undergraduate Texts in Mathematics. Springer, 2015. ISBN: 978-1-4939-2711-1. DOI: 10.1007/978-1-4939-2712-8.
- [SS03] Stein and Shakarchi. *Fourier Analysis*. Princeton University Press, 2003. ISBN: 978-0-691-11384-5.